

SEOHYEON CHA PH.D. STUDENT

SUMMARY

My research develops **efficient, trustworthy AI** for **decentralized, resource-constrained** systems. I'm interested in designing methods that preserve reliability under tight compute/memory/latency budgets, device heterogeneity, and non-stationary deployments. Recent directions include LLM quantization for efficient inference and continual learning in collaborative, privacy-preserving environments.

EDUCATION

The University of Texas at Austin	Austin, TX
<i>Ph.D. in Electrical and Computer Engineering (GPA: 4.0/4.0)</i>	2024 - Present
• Advisor: Prof. Haris Vikalo	
Korea Advanced Institute of Science and Technology (KAIST)	Daejeon, Korea
<i>M.S. in Electrical Engineering (GPA: 4.17/4.3)</i>	2022 - 2024
• Advisor: Prof. Joonhyuk Kang	
<i>B.S. in Electrical Engineering (GPA: 4.03/4.3, Summa Cum Laude)</i>	2017 - 2022

PUBLICATIONS

Highlights

1. S Cha, H Chen, D Kim, H Zhang, K Chan, G Veciana, H Vikalo, “[CoreQ: Learning-Free Mismatch Correction and Successive Rounding for Quantization](#),” submitted, 2026.
2. S Cha*, H Chen*, and H Vikalo, “[FedProTIP: Task-Agnostic Federated Continual Learning via Replay-Free Gradient Projection](#),” in *Transactions on Machine Learning Research (TMLR)*, 2026.
3. S Cha, K Chan, Ge Veciana, and H Vikalo, “[Joint Model Onloading and Offloading for Hierarchical Multi-Task Inference](#),” in *IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, 2026.
4. H Kang*, S Cha*, and J Kang, “[GeFL: Model-Agnostic Federated Learning with Generative Models](#),” in *IEEE Transactions on Mobile Computing*, 2025.
5. H Kang, S Cha, J Shin, J Lee, and J Kang, “[NeFL: Nested Federated Learning for Heterogeneous Clients](#),” in *IEEE Transactions on Mobile Computing*, 2025.

Additional Publications

1. D Kim, S Cha, H Chen, C Wang, H Vikalo, “[Quantized Gradient Projection for Memory-Efficient Continual Learning](#),” in *International Conference on Learning Representations (ICLR)*, 2026.
2. H Zhang, D Kim, S Cha, H Vikalo, “[FedRot-LoRA: Mitigating Rotational Misalignment in Federated LoRA](#),” in *International Conference on Machine Learning (ICML)*, 2026.
3. H Zhang, S Cha, HB Beytur, K Chan, G Veciana, H Vikalo, “[Online Learning for Multi-Layer Hierarchical Inference under Partial and Policy-Dependent Feedback](#),” arXiv preprint, 2026.

4. S Cha, H Kang, and J Kang, “[On the Temperature of Bayesian Graph Neural Networks for Conformal Prediction](#),” in *NeurIPS 2023 Workshop: New Frontiers in Graph Learning*, 2023.
5. S Cha, S Kim, J Seo, and J Kang, “[Intelligent Surface-aided Transmit-array Antenna in mmWave Communication System with Historical Channel Observation](#),” in *IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, 2022.

WORK EXPERIENCES	Kilby Labs (Embedded AI Team), Texas Instruments	Summer 2026
	Research Assistant, UT Austin	2024 - 2025
	Undergraduate Summer Internship, SK Hynix	Summer 2019
HONORS & AWARDS	• Best Project Award, ML on Real World Networks, UT Austin <i>Project: Distributed Continual Learning using Gradient Projection</i>	Fall 2024
	• National Science & Engineering Scholarship, Academic Excellence	2019 - 2021
	• Korean Governmental Scholarship, KAIST Graduate	2022 - 2024
	• Korean Governmental Scholarship, KAIST Undergraduate	2017 - 2018
TEACHING EXPERIENCES	Undergraduate Individual Study Assistant, KAIST	2023
	Teaching Assistant, KAIST	
	• EE205 Data Structures and Algorithms for Electrical Engineering, Fall 2022 • EE966 M.S. Seminar <Colloquium>, Spring/Fall 2023	
	Counseling Assistant, KAIST	Sep 2022 - Feb 2023
	• Counseled undergraduate/graduate students	
	Freshman Tutoring, KAIST	2018 - 2019
	• MAS101 Calculus 1, MAS102 Calculus 2	
SKILLS	Languages: English (Fluent), Korean (Native) Programming: , Python, MATLAB, $\text{\LaTeX}$, C++ Tools: PyTorch, HuggingFace, Linux, Git, Pandas	